Title of the Project

A Text spam and Ham classification End-to-end ML pipeline.

Introduction

The present world of digital communication is a massive segment of unsolicited messages that can be conveniently described as Spam in which the spam is a significant menace to the users in terms of security as well as productivity. The data science project being discussed focuses on developing a machine learning system that will be able to identify the presence or absence of spam in any piece of text. Conversely, "Ham" means clean/ unaltered legitimate messages whereas Spam as you can imagine is plain text usually employed in ill intentions.

This is because in our day-to-day lives, spam messages are commonly experienced in one form or another, e.g. emails or SMS. Since it is impossible to filter all the data manually due to the amount of data, the present project seeks to create a dynamic machine learning system. Through the history text data, the system will be able to identify and categorize messages as spam and ham with higher accuracy and ensure a scalable and effective solution to the issue of digital noise.

Problem Statement:

This data science project will construct a machine learning system that will be capable of identifying whether a text is spam or not. Spam refers to the texts that are mostly used when it is a spam. Ham refers to text that is not a spam and it is not dirty. We tend to come across the spam messages in our real life either through mails or sms. This project is a system that will be created to identify the spam and ham texts and in the process will be classifying them as spam and ham.

Key challenges include:

Security and Productivity Threats: Unsolicited messages are ever a source of threat to user data protection and overall productivity.

Complexity of Interaction Variable: Spam detection is done by detecting the complicated patterns in an array of text-based data where the surrounding of words define a clean message (Ham) or Spam.

Filtering Limitations of the Static Filters: Static filters which are built on traditional rules have no way to adapt to the new spam tricks, and as a result, one is forced to create a dynamic machine learning style which gets to learn through historical data.

Real-time Classification: A significant technical challenge is creating a system capable of handling real-time messages and delivering real-time classification responses through an API.

Scalability and Reliability: The ability to manage huge volumes of data and make the system reliable and scalable in various environments depends on such sophisticated infrastructure as cloud databases and containerization.

Objective and Scope:

Primary Goal: The development of an end-to-end machine learning operations pipeline that classifies and detects text using spam or ham.

Specific Goals

Dynamic Prediction: The objective of this is to come up with a machine learning architecture that is capable of predicting types of texts depending on past text data.
Automated Pipeline: Develop an automated system that consists of data ingestion and transformation into MongoDB Atlas and model training.
Web-Based API: API in fast mode or Flask interface to real-time prediction and training.
Scalable Infrastructure: Docker and AWS (EC2, S3, ECR) should be used so that the system can be ready to production.

Methodology

Information Ingestion: Fetches information in MongoDB Atlas and gets it ready to process it. Data validation Data drift and schema checks are used to guarantee model reliability.
Data Transformation: Applies NLP algorithms, such as Tokenization, Lemmatization and Vectorization (TF-IDF or Bag of Words).
Model Training: Plots several algorithms (MultinomialNB, GaussianNB, SVC) and hyperparameter optimisation is performed with the help of the grid search.
Model Evaluation: It applies accuracy, precision, recall and F1-Score to select the optimal model.
Deployment: Puts the completed model to production through AWS S3 and ECR through Docker.

Hardware and Software Requirements.

Language: Python 3.7+.
Web Framework: Fast API / Flask.
Machine Learning: Scikit-learn (GridSearchCV to optimize).
Database: MongoDB Atlas.
DevOps: Docker (Containerization) and GitHub Actions (CI/CD).
Cloud Infrastructure: AWS EC2, S3 and ECR.

Application/ Future Scope:

Application: Real-time Communication Security - Helping organisations to check the internal communication channels to avoid the spread of malicious links and keep their data intact.

Further Development: Deep Learning Retraining - Adding more sophisticated models such as LSTMs or Transformers (BERT) to get improved performance by having the capability of learning the context and sentiment of the word.

Project Timeline (Gantt Chart)